Project 04 — Personal Portfolio (Capstone)

Synthesizing the Full Web Development Stack

Summary: This document defines the capstone subject matter for Project 04: Designing, building, and publishing an authentic, responsive personal developer portfolio from scratch using semantic HTML, modern CSS, dynamic JavaScript DOM manipulation, and Git version control.

Version: 1.0

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Chapter I: Instructions

- These exercises are carefully arranged in order of difficulty, from easiest to hardest. We will not consider a successfully completed harder exercise if an easier one is not fully functional.
- Ensure that you have the appropriate permissions and correct directory structure for all your files.
- You must follow the exact submission procedures and naming conventions for every exercise.
- Your exercises will be checked and graded by your fellow classmates during peer evaluations.
- Additionally, your exercises may be validated by automated evaluation scripts. The evaluation is rigorous and strict. To avoid unpleasant surprises, be as thorough as possible.
- You must not leave any extraneous files in your directories other than those explicitly specified in the subject.
- Have a question? Ask the peer on your right. If not, try the peer on your left.
- Your reference guides are official documentation (MDN, man, Git docs) and verified technical resources.
- Examine the examples and edge cases carefully. They contain critical details.

— Did you check with your right-side neighbor? —

Chapter II: AI Instructions

The Web Piscine is foundational. Your primary mission is to build solid problem-solving reflexes, genuine technical autonomy, and collaborative peer learning.

In the modern tech ecosystem, generative AI shortcuts are tempting. However, shortcuts prevent the formation of neural connections required for deep engineering intuition.

Key Rules on AI Usage:

- Apply your own reasoning to each exercise before consulting external AI tools.
- Never ask an AI to generate solutions directly for you. You must write every line of code yourself.
- During peer evaluation and exams, you will have to explain every architectural decision and command. If you cannot explain your code, you will receive zero points.
- Good practice: You are confused about a concept like Flexbox or Git staging. You discuss it with your peer for 10 minutes, sketch it on paper, and it clicks.
- Bad practice: Copying code without understanding. You fail the peer defense because you cannot answer conceptual questions.

— Learning is not knowing the answer; it is developing the ability to find it. —

Chapter III: Foreword

"The only way to learn a new programming language is by writing programs in it." — Dennis Ritchie

You have learned Linux and Git to manage your workspace. You have learned HTML to give structure, CSS to give beauty, and JavaScript to give interactivity.

Now, bring all these disciplines together to build your own personal portfolio website from scratch.

— Your portfolio is proof of your autonomy and craftsmanship. —

Chapter IV — Exercise 00: Personal Portfolio

Directory: ex00/

Files to Submit: index.html, style.css, script.js, assets/, README.md

Authorized: HTML5, CSS3, JavaScript ES6+, Git

Your authentic digital presence.

- Build an authentic, responsive personal portfolio website from scratch without frameworks, templates, or UI libraries.
- Required sections: Header/Navbar, About Me with photo, Skills grid, Project Showcase (covering Projects 00-03), Hobbies, Contact Form, and Footer.
- Mandatory JavaScript Feature: Implement an interactive background color or theme switcher button that modifies styles dynamically on user click.
- The student must explain the complete event lifecycle: User action -> Event firing -> JavaScript callback -> DOM/CSS mutation.
- Implement mobile-responsive styling using Flexbox and CSS media queries.

! Templates (Bootstrap, Tailwind, WordPress) will result in an immediate 0.

— Craft every line of code with your own hands. —

Chapter V — Bounce 01: Profile Photo Switcher

Directory: bounce01/

Files to Submit: index.html, style.css, script.js, assets/

Authorized: DOM manipulation, element.src, element.alt

Interactive avatar customization.

- Implement an interactive button or avatar selector that switches the displayed profile photo between two or more images (e.g. casual vs professional).
- Use JavaScript DOM manipulation to update the image .src and .alt attributes dynamically with a smooth CSS transition.

! The page must not reload when switching photos.

— Small visual touches delight users. —

Chapter VI — Bounce 02: Language Switcher

Directory: bounce02/

Files to Submit: index.html, style.css, script.js, assets/

Authorized: JavaScript Objects, DOM textContent, dir="rtl"

Internationalization and bidirectional layout.

- Add a language switcher button that toggles the entire portfolio between English and Arabic.
- Organize translated strings using a clean JavaScript translation dictionary.
- Dynamically update the document direction: set dir="rtl" and lang="ar" for Arabic, and dir="ltr" for English.
- Ensure your CSS layout adapts smoothly to Right-to-Left (RTL) reading order without breaking alignment.

— A truly global application speaks multiple languages. —

Chapter XIV: Submission & Evaluation

- Submit your assignment to your Git repository as instructed. Only the work pushed to your repository branch will be evaluated during defense.
- Double-check your directory structure, file permissions, and exact file names before final submission.

! You must submit only the files explicitly required by the project instructions.

- Prepare to demonstrate your code live, explain your logic line-by-line, and answer questions about core concepts.

— RTFM and verify your work before peer defense! —